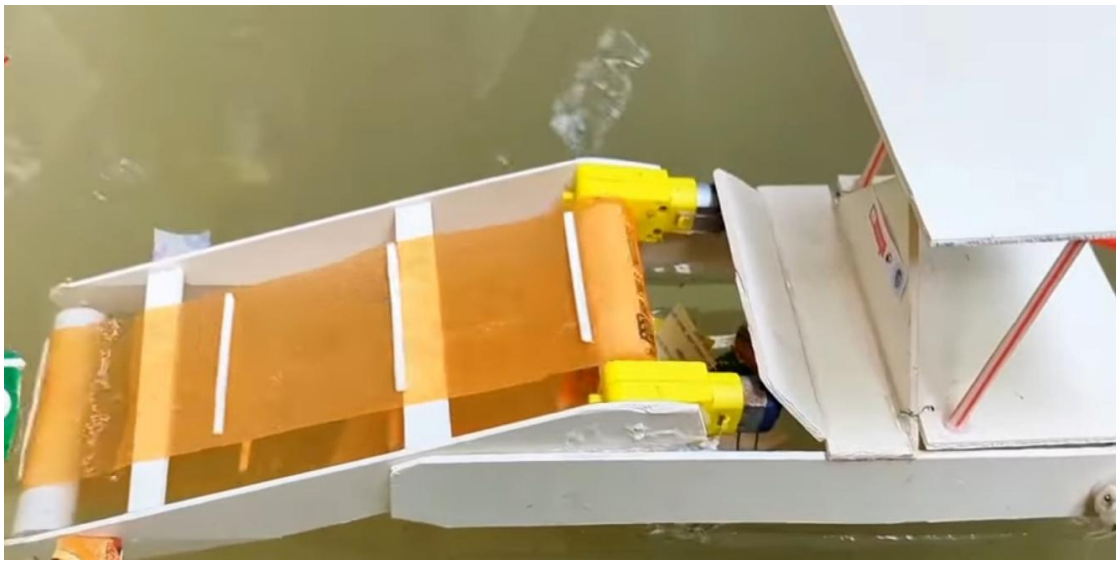
INDIAN INSTITUTE OF TECHNOLOGY, ROPAR

MECHANICAL ENGINEERIING DEPARTMENT

CP301 DEP PROJECT REPORT

Smart Water Trash Collecting Boat



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Smart Water Trash Collecting Boat

1. Introduction:

Water pollution has become a serious environmental problem due to the increasing amount of floating plastic and solid waste in rivers and lakes.

Traditional manual cleaning methods are inefficient and require continuous human effort. To solve this issue, the project proposes the development of a Miniature Autonomous River Cleaning Boat.



The boat will:

- Collect floating waste from water.
- Use solar energy for power.
- Use smart control systems for operation.

This project integrates:

- Mechanical design
- Electronics
- Automation

To create a sustainable and eco-friendly solution for cleaning water bodies.

2. Concepts:

I. Environmental Concern:

Rivers are heavily polluted with plastic waste which harms:

- Water quality
- Water life
- Ecosystems



II. Technological Approach:

The system uses:

- ESP32-cam module
- DC motors
- Sensors (IR sensor, Ultrasonic sensor)
- Automation systems

To operate the cleaning boat efficiently.

III. Sustainable solution:

The boat is designed to be

- Low-cost
- Eco-friendly
- Energy efficient
- Autonomous

So that it can clean rivers without continuous human involvement.

3. Problem Statement:

Rivers and lakes are becoming increasingly polluted due to the accumulation of floating plastic waste, bottles and other solid garbage. This pollution negatively affects aquatic ecosystems and reduces water quality. Currently, many cleaning activities rely on manual labour using nets and small boats, which is slow, risky and labour-intensive. Although some industrial cleaning machines exist, they are often expensive and not suitable for small water bodies such as local rivers, canals or ponds. Therefore, there is a need to design a low-cost, energy-efficient autonomous river cleaning boat capable of collecting floating waste effectively without constant human intervention. Such a system a can help improve water cleanliness and contributes to sustainable environment management.



4. Major Problems Identified:

a) Growing Water Pollution:

Plastic bottles and garbage in rivers are increasing rapidly, which harms aquatic life and water quality.

b) Limitations of Current Cleaning Methods:

Manual cleaning is slow, risky, labour-intensive process. large machines are costlynot suitable for smaller water bodies.

c) Need for autonomous Solution:

A solar-powered autonomous boat is required that can efficiently collect floating waste without human supervision.

5. Development of River Cleaning Machines (Evolution):

a) Manual Cleaning (Early stage):

Cleaning was done using nets and small boats. Problems were that it required heavy human effort, very slow process, unsafe for workers, and not suitable for highly polluted areas.



b) Remote Controlled Cleaning Boats:

It has features like the boats controlled by human operators, electric motors used for movement, able to collect floating waste. But it also has some limitations such as it still required human supervision.

c) Semi-autonomous Systems:

In this some additional components are added like ultrasonic sensors and GPS systems. It has features like basic obstacle detection, solar panels introduced, improved waste collection mechanisms.

d) Fully autonomous Solar Cleaning Boats (current Technology)

Modern systems include:

- AI-based waste detection
- GPS navigation
- Path planning algorithms
- Solar powered battery systems
- Automatic conveyor belt for waste collection

6. Reference Models:







7. Components Required:

a. Mechanical and Electrical Components used:

- Arduino UNO
- Battery
- IR Sensor
- Propellers
- DC Gear Motors
- Ultrasonic Sensor
- Motor Driver
- ESP32 – CAM module
- Conveyor Belt
- PVC Pipes





b. Electronics and Structural Components:

- ESP32 Camera Module
- Stainless steel Mesh Basket/ Storage Box
- Sunboard sheet
- Wiring and connectors

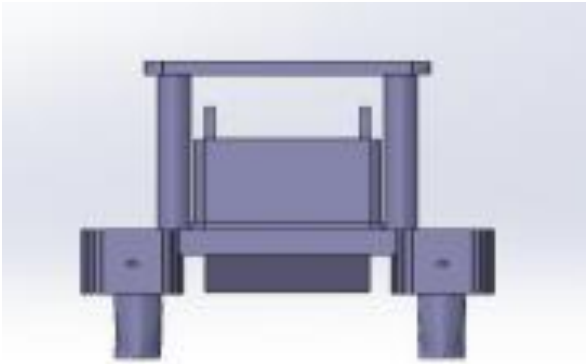


8. CAD Modelling and Structural Analysis:

a. CAD Modelling

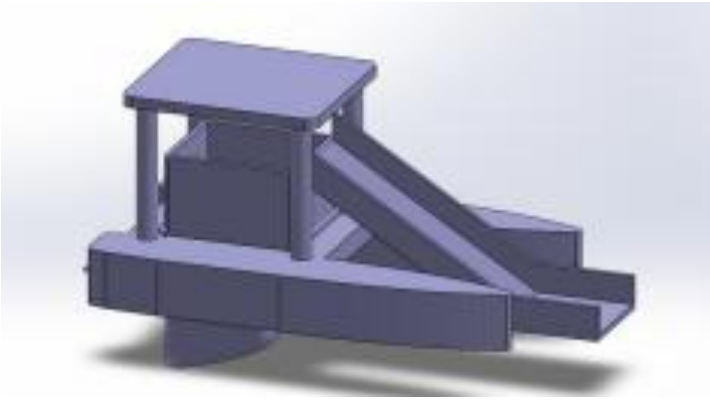
The complete boat structure was designed using SolidWorks software. The CAD model included:

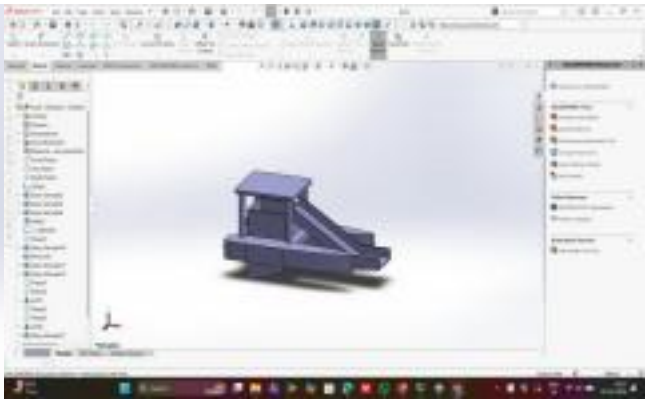
- Catamaran hull structure
- Conveyor support structure
- Sensor mounting positions
- Electronics compartment
- Propeller mounting arrangement



The catamaran hull configuration was selected to improve:

- Stability
- Load distribution
- Floating capability





b. Abaqus Structural Analysis

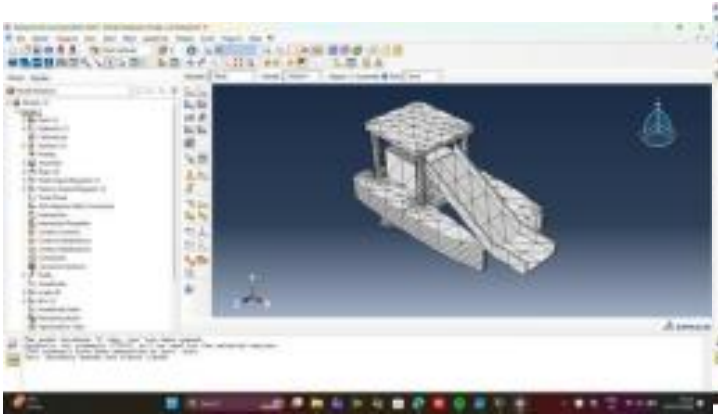
The CAD model was imported into Abaqus software for structural analysis.

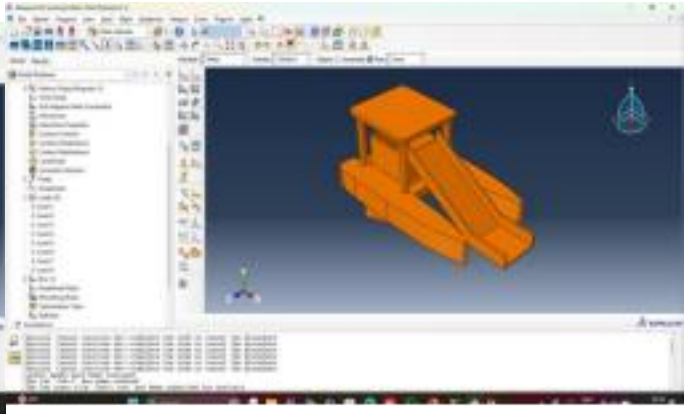
The following loads and conditions were considered:

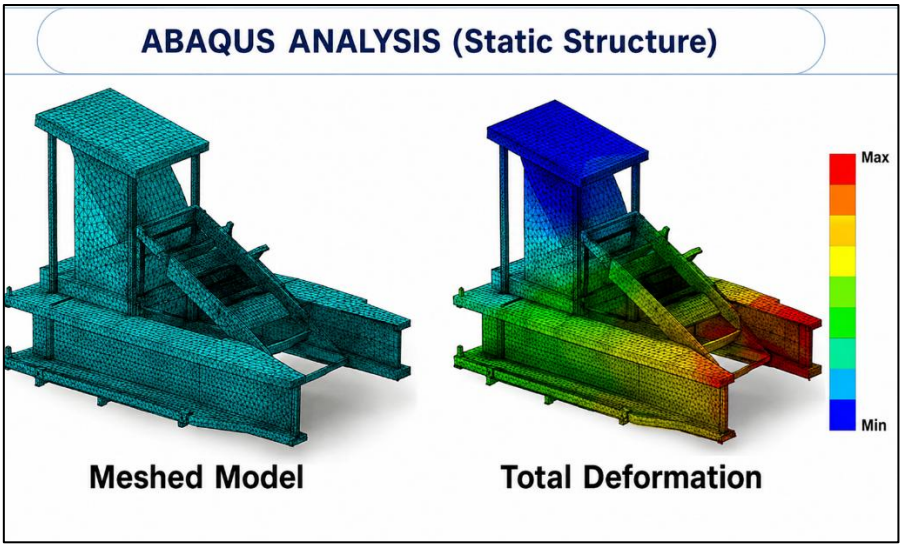
- Gravity loading
- Water pressure
- Operational loading
- Drag forces

Meshing was performed to evaluate structural deformation and stress distribution under loading conditions.

The analysis confirmed that the structure can withstand operational conditions with acceptable deformation limits.







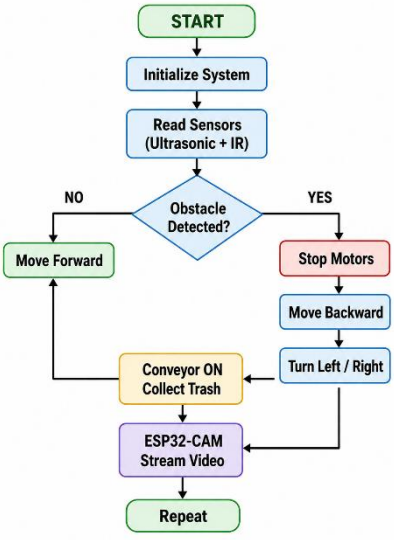
<u>Force Calculations</u>: we also calculated expected forces during Abaqus simulation.

Load Type	Value	Where to Apply
Gravity	9.81m/ s^2	Entire model
Garbage	2000Pa	Basket bottom
Water Pressure	1500Pa	Bottom hull
Drag Force	40N	Front Surface
Motor Thrust	80N	Motor mount
Conveyor Torque	5Nm	Conveyor Shaft

9. <u>System Architecture and Workflow:</u>

The smart autonomous trash collecting boat consists of three major subsystems:

1. Navigation and propulsion system
2. Sensing and obstacle detection system
3. Waste collection mechanism



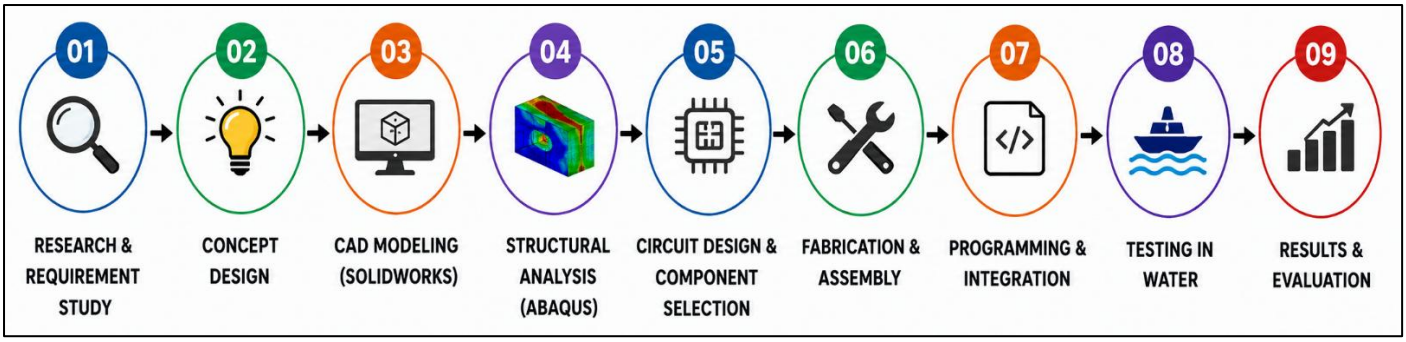
The propulsion system uses two DC motors connected to propellers for movement in water. The Arduino UNO acts as the central controller and controls all sensors and motors through programmed logic.

The ultrasonic sensor continuously scans the front area of the boat to detect obstacles such as walls, rocks or other floating objects. When an obstacle is detected within a predefined distance, the Arduino changes the direction of the boat automatically.

The IR sensor is positioned near the conveyor intake section to detect nearby floating waste. Once trash is detected, the conveyor mechanism can be activated for collection.

The ESP32-CAM module is used for wireless monitoring and live video streaming. This helps in observing the movement and surroundings of the boat remotely.

The conveyor belt mechanism lifts floating waste from the water surface and transfers it into the storage compartment.

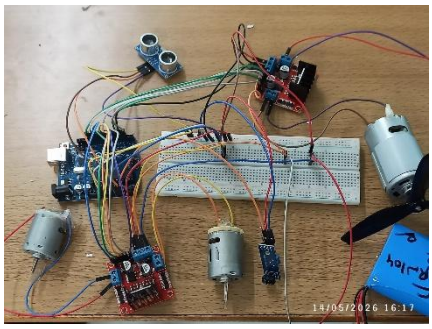


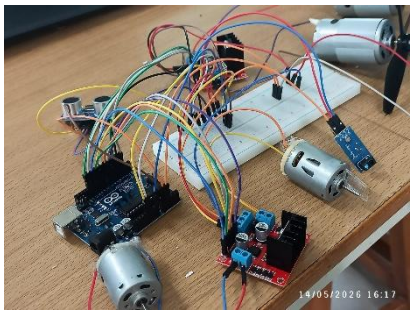
10.Embedded System and Electronics Integration:

The embedded system of the project integrates sensors, motor drivers and communication modules to achieve autonomous operation.

The Arduino UNO is used as the primary microcontroller for controlling:

- DC motors
- Conveyor mechanism
- Ultrasonic sensor
- IR sensor





The ESP32-CAM module provides wireless monitoring and future support for smart object detection.

The L298N motor driver controls the propulsion motors using PWM signals generated from the Arduino. Different combinations of motor rotation are used for forward movement, turning and reverse motion.

The ultrasonic sensor provides distance measurements for obstacle avoidance, while the IR sensor performs short-range object sensing near the conveyor intake area.

The system is powered using a 12V battery and a buck converter is used for voltage regulation of electronic components.

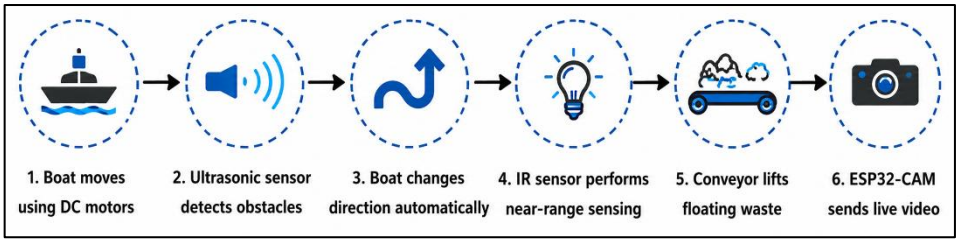
11. Sensor Working and Logic:

a. Ultrasonic Sensor Logic

The HC-SR04 ultrasonic sensor works by transmitting ultrasonic waves and receiving the reflected signals from nearby objects. The distance is calculated based on the time taken by the wave to return.

The ultrasonic sensor is used for:

- Obstacle detection
- River bank avoidance
- Collision prevention



If the measured distance becomes less than the threshold value, the Arduino changes the direction of the boat automatically.

b. IR Sensor Logic

The IR sensor is placed near the conveyor intake region and is used for short-range object detection.

The sensor detects nearby floating waste entering the collection region. This can later be used to activate the conveyor belt mechanism automatically.

12. Motor Control and Navigation:

Two DC motors connected with propellers are used for boat propulsion. The motors are controlled using an L298N motor driver module.

Different motor combinations are used for navigation:

- Both motors forward → Forward motion
- One motor forward and one reverse → Turning
- Both motors reverse → Backward motion

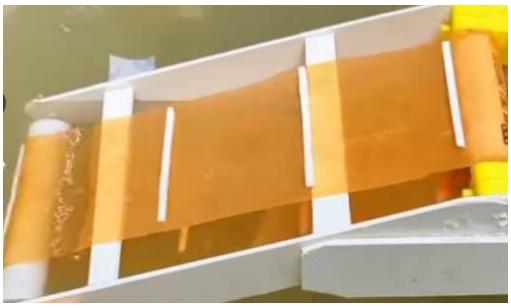


PWM speed control is implemented using the Arduino UNO for smooth movement control.

The navigation system combines sensor inputs and programmed logic to allow semi-autonomous movement of the boat.

13. Conveyor Belt Mechanism

The conveyor belt mechanism is designed to lift floating waste from the water surface into the storage compartment.



The conveyor is placed at the front section of the boat where floating waste first enters the intake area. A DC geared motor drives the conveyor mechanism.

The inclined conveyor structure helps in lifting waste efficiently while allowing water to drain back into the river.

This mechanism improves waste collection efficiency compared to manual net collection methods.

14. Testing and Experimental Results:

The prototype was tested in different stages to verify subsystem performance.

A. Motor Testing



The propulsion motors were tested for:

- Forward motion
- Reverse motion
- Turning operations

The motors responded successfully to PWM control signals from the Arduino.

B. Ultrasonic Sensor Testing



The ultrasonic sensor successfully detected nearby obstacles and produced distance values through the serial monitor.

Obstacle avoidance behaviour was achieved using programmed logic.

C. IR Sensor Testing

The IR sensor successfully detected nearby objects within short range and responded reliably during testing.

D. Integrated System Testing



The motors, ultrasonic sensor and IR sensor were integrated together to test autonomous navigation.

The boat successfully demonstrated:

- Forward movement
- Automatic obstacle avoidance
- Sensor-based directional changes

```
/*************************************************************
  AUTONOMOUS GARBAGE COLLECTING BOAT
  ------------------------------------------------------------
  CONNECTIONS USED IN THIS CODE

  MOTOR DRIVER 1
  D5  -> ENA1
  D6  -> IN1_1
  D7  -> IN2_1
  D9  -> IN3_1
  D10 -> IN4_1
  D11 -> ENB1

  MOTOR DRIVER 2
  D3  -> ENA2
  D4  -> IN1_2
  D8  -> IN2_2
  D12 -> IN3_2
  D13 -> IN4_2
  D2  -> ENB2

  ULTRASONIC SENSOR
  A0 -> TRIG
  A1 -> ECHO

  IR SENSOR
  A2 -> OUT
*************************************************************/

// ================= MOTOR DRIVER 1 =================
#define ENA1 5
#define IN1_1 6
#define IN2_1 7
#define IN3_1 9
```

```
#define IN4_1 10
#define ENB1 11

// ================= MOTOR DRIVER 2 =================
#define ENA2 3
#define IN1_2 4
#define IN2_2 8
#define IN3_2 12
#define IN4_2 13
#define ENB2 2

// ================= SENSORS =================
#define TRIG_PIN A0
#define ECHO_PIN A1
#define IR_PIN   A2

// ================= SETTINGS =================
const int MOTOR_SPEED = 220;
const int SAFE_DISTANCE = 20;      // cm
const int COLLECTION_TIME = 3000;  // ms

// ===================================================
void setup() {
  Serial.begin(9600);

  // Motor Driver 1 pins
  pinMode(ENA1, OUTPUT);
  pinMode(IN1_1, OUTPUT);
  pinMode(IN2_1, OUTPUT);
  pinMode(IN3_1, OUTPUT);
  pinMode(IN4_1, OUTPUT);
  pinMode(ENB1, OUTPUT);

  // Motor Driver 2 pins
  pinMode(ENA2, OUTPUT);
  pinMode(IN1_2, OUTPUT);
  pinMode(IN2_2, OUTPUT);
  pinMode(IN3_2, OUTPUT);
  pinMode(IN4_2, OUTPUT);
  pinMode(ENB2, OUTPUT);

  // Sensors
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(IR_PIN, INPUT);

  stopBoat();
  delay(1000);

  Serial.println("Garbage Collecting Boat Started");
}

// ===================================================
void loop() {
```

```
  long distance = getDistance();
  bool garbageDetected = (digitalRead(IR_PIN) == LOW);
  // If your IR sensor gives opposite behavior, change LOW to HIGH.

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm | Garbage: ");
  Serial.println(garbageDetected ? "YES" : "NO");

  // 1. Garbage detected
  if (garbageDetected) {
    stopBoat();
    Serial.println("Garbage detected - collecting...");
    delay(COLLECTION_TIME);
  }

  // 2. Obstacle detected
  else if (distance > 0 && distance < SAFE_DISTANCE) {
    Serial.println("Obstacle detected - turning right");

    stopBoat();
    delay(500);

    turnRight();
    delay(1200);

    stopBoat();
    delay(300);
  }

  // 3. Normal movement
  else {
    moveForward();
  }

  delay(100);
}

// ================================================
long getDistance() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH, 30000);

  if (duration == 0) {
    return -1;
  }

  return duration * 0.034 / 2;
```

```
}

// ================================================
void setSpeed(int speedValue) {
  analogWrite(ENA1, speedValue);
  analogWrite(ENB1, speedValue);
  analogWrite(ENA2, speedValue);
  analogWrite(ENB2, speedValue);
}

// ================================================
void moveForward() {
  setSpeed(MOTOR_SPEED);

  // Driver 1
  digitalWrite(IN1_1, HIGH);
  digitalWrite(IN2_1, LOW);
  digitalWrite(IN3_1, HIGH);
  digitalWrite(IN4_1, LOW);

  // Driver 2
  digitalWrite(IN1_2, HIGH);
  digitalWrite(IN2_2, LOW);
  digitalWrite(IN3_2, HIGH);
  digitalWrite(IN4_2, LOW);
}

// ================================================
void turnRight() {
  setSpeed(MOTOR_SPEED);

  // Left motors forward
  digitalWrite(IN1_1, HIGH);
  digitalWrite(IN2_1, LOW);
  digitalWrite(IN1_2, HIGH);
  digitalWrite(IN2_2, LOW);

  // Right motors backward
  digitalWrite(IN3_1, LOW);
  digitalWrite(IN4_1, HIGH);
  digitalWrite(IN3_2, LOW);
  digitalWrite(IN4_2, HIGH);
}

// ================================================
void stopBoat() {
  setSpeed(0);

  // Driver 1
  digitalWrite(IN1_1, LOW);
  digitalWrite(IN2_1, LOW);
  digitalWrite(IN3_1, LOW);
  digitalWrite(IN4_1, LOW);
```

```
  // Driver 2
  digitalWrite(IN1_2, LOW);
  digitalWrite(IN2_2, LOW);
  digitalWrite(IN3_2, LOW);
  digitalWrite(IN4_2, LOW);
}
```

15. Advantages of Proposed System:

- Low-cost and compact design
- Reduced manual cleaning effort
- Eco-friendly operation
- Lightweight structure
- Suitable for rivers, ponds and canals
- Semi-autonomous operation
- Easy fabrication and maintenance

16. Limitations:

- Limited battery backup
- Sensor inaccuracies under outdoor conditions
- ESP32-CAM connectivity issues
- Limited waste carrying capacity
- Sensitivity to water splashes and environmental conditions
- Basic autonomous navigation only

17. Future Scope:

Future improvements can include:

- AI-based trash identification using computer vision
- GPS-based navigation
- Solar-powered charging system
- Automatic path planning
- Mobile application control
- Real-time IoT monitoring
- Improved waterproof electronics housing
- Fully autonomous river cleaning operation



18. Conclusion:

The Smart Autonomous Trash Collecting Boat project demonstrates the use of embedded systems, sensors and mechanical design for solving water pollution problems.

The developed prototype successfully integrates propulsion, obstacle detection and waste collection mechanisms into a functional semi-autonomous system. The project demonstrates how low-cost automation can help improve water cleaning efficiency while reducing manual labour.

The project also provides a strong foundation for future implementation of AI-based waste detection and fully autonomous water cleaning systems.



19.Literature References:

We read some research papers based on this project:-

- https://ijaems.com/detail/solar-based-garbage-cleaning-boat/
- https://www.researchgate.net/publication/392178806_IOT_Controlled_Water_Trash_Collector_Boat
- https://www.ijitee.org/wp-content/uploads/papers/v9i6/F3752049620.pdf
- https://www.researchgate.net/publication/364118184_Design_and_Prototype_Development_of_Portable_Trash_Collector_Boat_for_Small_Stream_Application